

Document Technical Blueprint

1. Overview

The **Automated Employee Onboarding & Offboarding System** is designed using a modular architecture that integrates employee management, workflow automation, document management, asset tracking, and reporting. The system enables secure and efficient management of the complete employee lifecycle while reducing manual intervention.

2. System Architecture

The system follows a three-tier architecture:

- **Presentation Layer:** Provides the user interface for HR, managers, IT staff, and administrators.
- **Application Layer:** Implements business logic, workflow automation, validation, notifications, and reporting.
- **Data Layer:** Stores employee information, documents, task details, asset records, and audit logs in a centralized database.

3. Technology Stack

Component	Technology
Programming Language	Python
Frontend	Tkinter / HTML, CSS, JavaScript
Backend	Python
Database	SQLite / MySQL
ORM (Optional)	SQLAlchemy
Version Control	Git
Reporting	Python ReportLab / Excel Export
Notifications	SMTP Email API

4. System Modules

Employee Management

- Employee registration
- Employee profile management

- Department assignment
- Status management

Onboarding Module

- New employee registration
- Document submission
- Approval workflow
- Task assignment
- Account creation

Offboarding Module

- Exit request processing
- Clearance workflow
- Asset return tracking
- Access revocation
- Final documentation

Document Management

- Secure document upload
- Document verification
- Document storage
- Download and retrieval

Workflow Engine

- Automated task generation
- Multi-level approvals
- Workflow status tracking
- Reminder notifications

Asset Management

- Asset allocation
- Asset return verification
- Asset inventory updates

Reporting Module

- Employee reports
- Onboarding reports
- Offboarding reports
- Department statistics
- Audit reports

5. Database Design

Employee Table

- Employee_ID (Primary Key)
- Name
- Email
- Phone
- Department
- Designation
- Joining_Date
- Status

Documents Table

- Document_ID
- Employee_ID
- Document_Name
- Upload_Date
- Verification_Status

Tasks Table

- Task_ID
- Employee_ID
- Assigned_To
- Task_Name
- Due_Date

- Status

Assets Table

- Asset_ID
- Employee_ID
- Asset_Name
- Asset_Type
- Allocation_Date
- Return_Status

Users Table

- User_ID
- Username
- Password (hashed)
- Role
- Last_Login

6. Workflow

Onboarding Process

1. HR registers a new employee.
2. Employee receives onboarding instructions.
3. Required documents are uploaded and verified.
4. IT creates user accounts and grants system access.
5. Assets are allocated.
6. Manager completes induction tasks.
7. Employee status changes to **Active**.

Offboarding Process

1. HR initiates the exit process.
2. Manager approves the request.
3. IT disables system access.
4. Company assets are returned.

5. Exit clearance is completed.
6. Employee status changes to **Inactive**.
7. Records are archived.

7. Security Features

- User authentication
- Role-based access control (RBAC)
- Password hashing
- Input validation
- Secure document storage
- Activity logging
- Data backup and recovery

8. Performance Considerations

- Optimized database queries
- Efficient workflow execution
- Centralized data management
- Scalable modular architecture
- Fast report generation

9. Deployment Environment

- Operating System: Windows / Linux
- Python 3.x Runtime
- SQLite or MySQL Database Server
- Local or Cloud Deployment
- Modern Web Browser (for web-based interface)

10. Future Enhancements

- Single Sign-On (SSO) integration
- AI-assisted onboarding recommendations
- Biometric attendance integration
- Mobile application support

- Cloud-based deployment
- HR analytics dashboards
- Integration with payroll and ERP systems

Summary

The technical blueprint defines the architecture, modules, database structure, workflows, and security mechanisms of the Automated Employee Onboarding & Offboarding System. The modular design supports scalability, maintainability, and secure data management while enabling efficient automation of employee lifecycle processes.